
Calibrated Uncertainty for Vision-Language Models: Bayesian Approximations to Reduce Overconfident Hallucinations in VQA

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Abstract

Vision-language models (VLMs) achieve strong performance on visual question answering (VQA) but often produce *overconfident* incorrect answers, a common form of hallucination that undermines trustworthiness. This project studies uncertainty quantification and confidence calibration for VLMs, with an emphasis on enabling reliable abstention (e.g., “I don’t know”) under uncertainty. We propose a lightweight Bayesian treatment of a pretrained VLM using Laplace approximation on the output head to capture epistemic uncertainty without large-scale retraining. We evaluate calibration and selective prediction on VQA benchmarks using Expected Calibration Error (ECE), negative log-likelihood, reliability diagrams, and risk–coverage curves. Our goal is to determine whether Bayesian approximations improve confidence reliability and reduce high-confidence errors relative to strong post-hoc and approximate Bayesian baselines.

1 Introduction

Vision-language models (VLMs) integrate visual and textual representations to solve multimodal tasks such as image captioning and visual question answering (VQA). Despite impressive accuracy, VLMs frequently output incorrect answers with high confidence, especially when questions are ambiguous, visual evidence is weak, or inputs are out-of-distribution. This overconfidence is problematic in downstream decision-making: a model that is *wrong but certain* is often more dangerous than one that is unsure.

A principled way to address overconfident hallucinations is to improve the *calibration* of model confidence. Calibration asks whether a predicted confidence (e.g., 0.9) corresponds to an empirical correctness frequency (about 90%). Deep networks are known to be miscalibrated [5], and the situation can be more complex in multimodal settings due to distribution shifts and entangled visual-textual uncertainty.

This project investigates **calibrated uncertainty** for VLMs. We focus on three questions: (i) how miscalibrated are common pretrained VLMs on VQA, (ii) can Bayesian approximations (Laplace) improve uncertainty estimates and reduce overconfident errors, and (iii) does improved uncertainty lead to better *selective prediction*, where the model abstains when unsure.

Contributions (planned). We propose (1) a controlled evaluation protocol for VLM calibration on VQA, including abstention; (2) a lightweight Laplace approximation applied to the VLM output head to model epistemic uncertainty; and (3) a systematic comparison against strong baselines (temperature scaling, MC dropout, ensembles) under limited compute.

2 Methodology

2.1 Task and model

We consider VQA, where each example consists of an image x and question q , and the model predicts an answer y from a candidate set $\mathcal{Y}$. We start from a publicly available pretrained VLM (e.g., BLIP-2 or a comparable open-source model), producing logits $\mathbf{z}(x, q) \in \mathbb{R}^{|\mathcal{Y}|}$ and probabilities

$$p(y \mid x, q) = \text{softmax}(\mathbf{z}(x, q))_y. \quad (1)$$

We define the model confidence as $\max_{y \in \mathcal{Y}} p(y \mid x, q)$.

2.2 Calibration evaluation

We evaluate calibration using:

- **Expected Calibration Error (ECE)** [5]: bin predictions by confidence and measure the discrepancy between average confidence and accuracy per bin.
- **Negative log-likelihood (NLL)**: assesses probabilistic sharpness and correctness.
- **Reliability diagrams**: visualize calibration gaps.

2.3 Selective prediction (“I don’t know”)

To study safe refusal, we use selective prediction [3]. Given a threshold τ , the model *abstains* if its confidence is below τ :

$$\hat{y}(x, q) = \begin{cases} \arg \max_y p(y \mid x, q), & \text{if } \max_y p(y \mid x, q) \geq \tau, \\ \text{“I don’t know”}, & \text{otherwise.} \end{cases}$$

We report **coverage** (fraction answered) and **selective risk** (error rate among answered), and summarize performance via **risk–coverage curves** and area-under-curve metrics.

2.4 Baselines

We compare against:

- **Temperature scaling** [5]: calibrate logits via $\mathbf{z}/T$ on a validation split.
- **MC dropout** [4]: approximate Bayesian inference by stochastic forward passes.
- **Deep ensembles** [6]: average predictions across independently trained heads/models (as compute permits).

2.5 Laplace approximation for VLM confidence

Exact Bayesian inference over VLM parameters is intractable. We adopt Laplace approximation as a scalable Bayesian method [2]. Let θ be parameters of the (frozen) backbone and w the parameters of the final answer head. We keep θ fixed and apply Laplace only to w for feasibility.

After fitting w (or lightly adapting it) to a VQA training split, we form a Gaussian approximation to the posterior:

$$p(w \mid \mathcal{D}) \approx \mathcal{N}(\hat{w}, \mathbf{H}^{-1}), \quad (2)$$

where $\hat{w}$ is the MAP estimate and $\mathbf{H}$ is a curvature approximation (e.g., diagonal or Kronecker-factored) of the negative log-posterior at $\hat{w}$.

Predictive probabilities marginalize parameters:

$$p(y \mid x, q, \mathcal{D}) = \int p(y \mid x, q, w) p(w \mid \mathcal{D}) dw, \quad (3)$$

approximated via Gaussian quadrature or Monte Carlo sampling over w . We then derive calibrated confidence from these Bayesian predictive probabilities and evaluate whether this reduces overconfident errors and improves abstention.

2.6 Experimental plan

Datasets: VQA v2 [1] and optionally OK-VQA [7] (higher ambiguity/knowledge demands). We will: (i) measure baseline miscalibration and hallucination-like overconfident errors, (ii) apply temperature scaling, MC dropout, ensembles, and Laplace-head, (iii) compare ECE/NLL and selective prediction performance, and (iv) analyze failure cases by question type and image difficulty.

3 Related Work

Prior work documents that large multimodal models can output answers not grounded in visual evidence, often with high confidence. This motivates uncertainty-aware inference and evaluation protocols focusing on reliability rather than accuracy alone. Guo et al. [5] showed that modern neural networks are often miscalibrated and popularized temperature scaling as a strong post-hoc baseline. Calibration under shift and in multimodal settings remains an active area, and VQA provides a natural testbed where correctness and confidence can be systematically measured. MC dropout [4] and deep ensembles [6] are widely used approximate methods for epistemic uncertainty. Laplace approximation offers a complementary approach by approximating the posterior around a MAP solution; recent work has made Laplace practical for deep networks [2]. Selective classification formalizes decision-making with abstention [3]. Risk–coverage curves provide a principled way to evaluate whether confidence estimates support safe refusal. We apply these tools to VLMs in VQA to measure how uncertainty calibration affects overconfident hallucinations.

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A GenAI Disclosure

Portions of this proposal were drafted with assistance from a generative AI tool for writing support and organization. All technical decisions, claims, and planned experiments were reviewed and are the responsibility of the authors.